

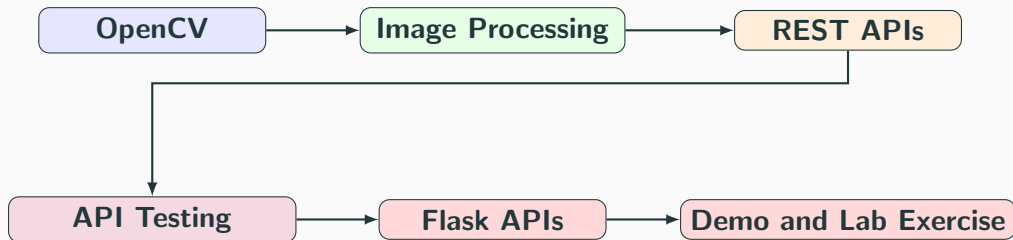
Computer Systems Design

Lab 2: Building Computer Vision APIs using OpenCV and Flask

Indian Institute of Technology Bhilai

5th of August, 2026

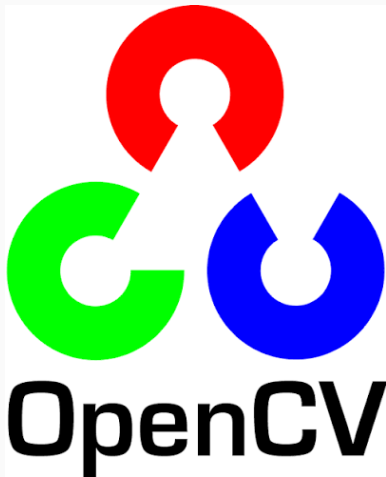
Today's Roadmap



OpenCV

What is OpenCV?

- Open Source Computer Vision Library
- Real-time image and video processing
- Widely used in AI, Robotics and Automation



Reading an Image

```
import cv2
img = cv2.imread("cat.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2.imshow("Gray", gray)
cv2.waitKey(0)
```

Note: Install OpenCV-python by running "pip install opencv-python"

A good PDF containing opencv commands in python is :

<https://depcnt.nitc.ac.in/electrical/ipg-new/python/opencv-python-tutroals.pdf>

Finding Contours

```
import cv2
img = cv2.imread("cat.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray,50,150)
cnts,_ = cv2.findContours(
    edges,
    cv2.RETR_EXTERNAL,
    cv2.CHAIN_APPROX_SIMPLE
)
cv2.drawContours(img,cnts,-1,(0,255,0),2)

cv2.imshow("Contours", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Useful OpenCV Functions for the Course Project

```
# Read and preprocess maps
cv2.imread()           # Read map/image
cv2.cvtColor()         # RGB -> Grayscale
cv2.resize()           # Standardize image size
# Image enhancement
cv2.GaussianBlur()     # Remove noise
cv2.threshold()        # Binary segmentation
cv2.Canny()            # Detect terrain boundaries
# Shape analysis
cv2.findContours()     # Extract contour boundaries
cv2.drawContours()     # Visualize regions
cv2.contourArea()      # Estimate pond/catchment area
cv2.arcLength()        # Boundary perimeter
cv2.boundingRect()     # Bounding box of a region
# Visualization
cv2.rectangle()        # Highlight candidate regions
cv2.putText()          # Display measurements
cv2.imwrite()          # Save processed results
```

REST APIs

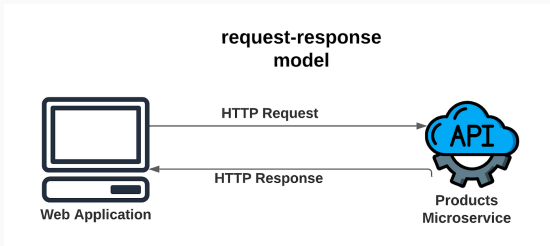
What is an API?

Application Programming Interface

An API exposes functionality of an application to other programs through well-defined endpoints.

Examples

- GitHub API
- Google Maps
- OpenWeather
- ChatGPT API
- OpenZenith



REST APIs

REST APIs expose resources using HTTP.

Method	Purpose	Example
GET	Read	Fetch Image
POST	Upload	Upload Image
PUT	Update	Update Metadata
DELETE	Remove	Delete Resource

Typical Response

- Status Code (200,404,500)
- Headers
- JSON / Image

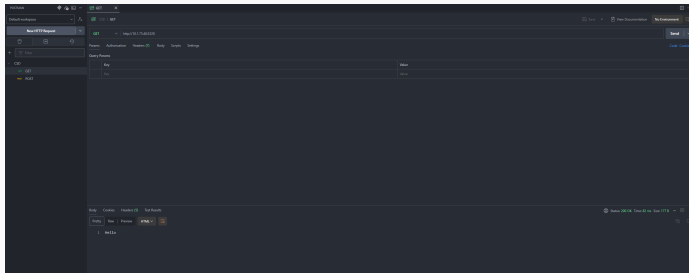
Post Man (VS Code)

VS Code extension /
Standalone App for testing
REST APIs.

Workflow

1. Select GET/POST
2. Enter URL
3. Add Headers/File
4. Send Request
5. Inspect Response

No Python code required.



From Public APIs to Our Own API

Try these APIs

```
GET https://catfact.ninja/fact
```

```
GET https://dog.ceo/api/  
breeds/image/random
```

```
GET https://api.github.com/  
users/octocat
```

Observe

- Status Code
- Headers
- JSON Response



Next, you'll replace the public API with your own Flask server.

Flask

What is Flask?

Flask

Flask is a lightweight Python web framework used to build web applications and REST APIs.

Why Flask?

- Lightweight and easy to learn
- Simple URL routing
- Handles HTTP requests and responses
- Easily integrates with OpenCV, NumPy, ML models
- Ideal for rapid API development

Minimal Flask API

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello"

app.run(host="0.0.0.0", port=5000)
```

Image Processing API

```
@app.post("/gray")
def gray():

    file = request.files["image"]

    img = cv2.imdecode(
        np.frombuffer(file.read(), np.uint8),
        cv2.IMREAD_COLOR)

    gray = cv2.cvtColor(img,
                        cv2.COLOR_BGR2GRAY)

    _, out = cv2.imencode(".jpg", gray)

    return Response(out.tobytes(),
                    mimetype="image/jpeg")
```


Available Application Ports

Each allocated container has a dedicated SSH port. The remaining application ports are derived from the SSH port number.

Service	Port Rule	Example (SSH = 2210)
SSH	2200	2210
Application 1	3000	3210
Application 2	4000	4210
Application 3	5000	5210
Application 4	6000	6210
Application 5	7000	7210

Example

If your assigned SSH port is 2210, then your application ports are 3210, 4210, 5210, 6210, and 7210.

Lab Exercise & Report Submission

Lab Exercise

1. SSH into your allocated system.
2. Implement a Flask application.
3. Create the following API routes:
 - /gray
 - /blur
 - /edges
 - /contours
4. Test each route using PostMan(or any other tool).
5. Verify the returned processed images.

Report Submission

- Student Name & Roll Number
- Assigned System i.e Hostname (stuXX-sysXX)
- Flask source code (in report itself, not .py files)
- PostMan screenshots
- Input & Output images
- Brief observations (if any)

Submission

Submit a single PDF report in Google Classroom/Form before EOD today.

Thank You!